

Olympiad Test Paper — Mathematics (Class 7)

Chapter 15: Finding the Unknown

Subject	Mathematics (NCERT Class 7)
Chapter	15 — Finding the Unknown (Linear Equations)
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Solve every equation by balancing / transposition and you may always check by substituting back. Most questions need two to four steps; watch for sign traps, bracket-distribution slips, and "average of speeds"-style shortcuts that don't work. Do not write on this paper — mark answers on your answer sheet.

1. The phrase "7 less than twice a number n " is written as:

- (A) $2(n - 7)$ (B) $2n - 7$
(C) $7 - 2n$ (D) $2n + 7$

2. "When 5 is added to thrice a number, the result is 26." As an equation this is:

- (A) $3(x + 5) = 26$ (B) $5x + 3 = 26$
(C) $3x - 5 = 26$ (D) $3x + 5 = 26$

3. If $4x - 9 = 2x + 7$, then the value of $3x + 1$ is:

- (A) 25 (B) 24
(C) 17 (D) 9

4. One-third of a number exceeds one-fourth of the same number by 5. The number is:

- (A) 60 (B) 5
(C) 20 (D) 12

5. Solve: $7x + 4 = 3x + 20$.

- (A) 6 (B) 2
(C) 4 (D) 3

6. A number increased by 8 equals three times the result of decreasing the same number by 4. The number is:

- (A) 8
(C) 14

- (B) 5
(D) 10

7. Solve: $3(2x + 1) = 27$.

- (A) 6
(C) 5

- (B) 9
(D) 4

8. Solve: $(2x - 1)/3 = 5$.

- (A) 7
(C) 16

- (B) 8
(D) 23

9. Which value of x satisfies $5x - 7 = 2x + 5$?

- (A) 2
(C) 6

- (B) 3
(D) 4

10. The sum of three consecutive integers is 72. The largest of them is:

- (A) 23
(C) 24

- (B) 25
(D) 21

11. A father is 4 times as old as his son, and the sum of their ages is 50 years. The son's age is:

- (A) 40
(C) 10

- (B) 12
(D) 8

12. Four pencils and a ₹10 sharpener together cost ₹38. The cost of one pencil (in ₹) is:

- (A) 7
(C) 12

- (B) 9
(D) 6

13. The length of a rectangle is 5 cm more than its breadth, and its perimeter is 50 cm. The breadth (in cm) is:

- (A) 10
(C) 20

- (B) 15
(D) 7.5

14. A matchstick pattern uses 4 sticks for 1 square, 7 for 2 squares, 10 for 3 squares (each extra square adds 3 sticks). How many squares can be made with exactly 31 sticks?

- (A) 7
(C) 10

- (B) 9
(D) 11

15. A number increased by 12 becomes three times the number. The number is:

- (A) 4
(C) 12

- (B) 6
(D) 3

16. Solve: $x/2 + x/3 = 10$.

- (A) 6
(C) 24

- (B) 4
(D) 12

17. Solve: $4(x - 1) = 2(x + 3)$.

(A) 2

(B) 7

(C) 1

(D) 5

18. The sum of a number and its successor is 5 less than three times the number. The number is:

(A) 5

(B) 4

(C) 7

(D) 6

19. The length of a rectangle is twice its breadth, and its perimeter is 54 cm. The length (in cm) is:

(A) 18

(B) 9

(C) 27

(D) 12

20. A box holds twenty ₹1 and ₹2 coins worth ₹35 in all. The number of ₹2 coins is:

(A) 5

(B) 15

(C) 10

(D) 20

21. For which equation is $x = 3$ the solution?

(A) $2x + 1 = 5$

(B) $x - 3 = 3$

(C) $4x - 5 = 7$

(D) $x/3 = 9$

22. Half of a number, decreased by 6, equals one-third of the same number. The number is:

(A) 12

(B) 18

(C) 9

(D) 36

23. Solve: $5 - 2(x - 1) = 1$.

(A) -3

(B) 3

(C) 2

(D) 4

24. Reena is 5 years older than Sam. In 3 years the sum of their ages will be 33. Sam's present age is:

(A) 11

(B) 8

(C) 16

(D) 14

25. In the sequence 3, 5, 7, 9, ... (nth term $2n + 1$), which term equals 25?

(A) the 11th

(B) the 12th

(C) the 13th

(D) the 24th

26. Solve: $3(x + 2) = 2(x + 5)$.

(A) 4

(B) 6

(C) 16

(D) 2

27. Solve: $(x - 3)/2 = (x + 1)/4$.

- (A) 5 (B) 3
(C) 11 (D) 7

28. ₹120 is shared between A and B so that A gets twice as much as B. B's share (in ₹) is:

- (A) 80 (B) 60
(C) 40 (D) 30

29. Find a number such that doubling it and adding 3 gives the same result as tripling it and subtracting 2.

- (A) 5 (B) 1
(C) -5 (D) 6

30. Two angles of a triangle are equal, and the third is 30° more than each of them. Each equal angle measures:

- (A) 30° (B) 80°
(C) 40° (D) 50°

31. Solve: $12 - 3x = 2x - 8$.

- (A) 2 (B) -4
(C) 4 (D) 5

32. "When a number is divided by 5 and 3 is added, the result is 8." The number is:

- (A) 40 (B) 25
(C) 55 (D) 15

33. In a two-digit number the units digit is 3 more than the tens digit, and the sum of the two digits is 11. The number is:

- (A) 47 (B) 74
(C) 38 (D) 29

34. Solve: $2x/3 - x/2 = 1$.

- (A) 3 (B) 2
(C) 6 (D) 12

35. A family bought 5 tickets for ₹155. Adult tickets cost ₹40 and child tickets ₹25. The number of child tickets bought was:

- (A) 3 (B) 2
(C) 5 (D) 4

36. $x = -2$ is a solution of which equation?

- (A) $3x + 1 = 5$ (B) $2x + 9 = 5$
(C) $x - 2 = 0$ (D) $4x = 8$

37. Solve: $2(3x - 1) = 4(x + 2)$.

- (A) 3 (B) 10
(C) 5 (D) 1

38. A taxi charges ₹50 fixed plus ₹12 per km. A ride costing ₹158 covered how many km?

- (A) 8 (B) 9
(C) 10 (D) 12

39. ₹90 is divided among three friends so that the second gets twice the first and the third gets three times the first. The third friend's share (in ₹) is:

- (A) 45 (B) 15
(C) 30 (D) 90

40. Solve: $(x - 1)/2 + (x + 1)/3 = 4$.

- (A) 4 (B) 6
(C) 5 (D) 7

41. Solve: $6x - 5 = 4x + 7$.

- (A) 1 (B) 6
(C) 3 (D) 11

42. The sum of two numbers is 30, and one is 6 more than the other. The larger number is:

- (A) 18 (B) 12
(C) 24 (D) 15

43. An isosceles triangle has two equal sides of length x and a base of length $(x - 2)$. Its perimeter is 25. Then x equals:

- (A) 7 (B) 11
(C) 9 (D) 8

44. In the pattern 5, 8, 11, 14, ... (each term is 3 more than the one before), which term equals 47?

- (A) the 16th (B) the 14th
(C) the 47th (D) the 15th

45. "The cost of n notebooks at ₹15 each, after a ₹20 discount on the bill" is written as:

- (A) $15n - 20$ (B) $15(n - 20)$
(C) $20 - 15n$ (D) $15n + 20$

46. A two-digit number has digit-sum 9, and its tens digit is twice its units digit. The number is:

- (A) 36 (B) 72
(C) 81 (D) 63

47. $x = 2$ is a solution of every equation below **except**:

(A) $3x - 1 = 5$

(B) $2x + 3 = 5$

(C) $x + 4 = 6$

(D) $5x = 10$

48. Solve: $3 - (2x - 5) = 4$.

(A) 4

(B) -2

(C) 2

(D) 6

49. ₹600 is shared among A, B and C so that B gets ₹50 more than A and C gets ₹50 more than B. A's share (in ₹) is:

(A) 200

(B) 250

(C) 150

(D) 100

50. Twice a number minus 7 equals the number plus 5. The number is:

(A) 6

(B) 12

(C) 2

(D) -12

51. Which value of x satisfies BOTH $2x - 1 = x + 3$ and $x/2 + 1 = 3$?

(A) 3

(B) 2

(C) 4

(D) 5

52. The angles of a triangle are in the ratio 1 : 2 : 3. The largest angle is:

(A) 30°

(B) 60°

(C) 120°

(D) 90°

53. Solve: $x/2 + 4 = x - 1$.

(A) 5

(B) 6

(C) 8

(D) 10

54. When 4 is subtracted from a number and the result is divided by 3, the answer is 5. The number is:

(A) 11

(B) 15

(C) 19

(D) 23

55. The sum of four consecutive integers is 90. The smallest of them is:

(A) 20

(B) 21

(C) 22

(D) 24

56. Solve: $2(x + 4) - 3 = x + 9$.

(A) 4

(B) 5

(C) 9

(D) 1

57. A mother is 26 years older than her daughter. In 6 years she will be 3 times as old as her daughter. The daughter's present age is:

(A) 7

(B) 5

(C) 8

(D) 10

58. A shopkeeper sold 30 fruits for ₹201, charging ₹8 per apple and ₹5 per orange. The number of apples sold was:

(A) 13

(B) 17

(C) 15

(D) 20

59. Solve: $(2x + 1)/3 = (x + 4)/2$.

(A) 5

(B) 7

(C) 8

(D) 10

60. Find the positive integer for which 3 times the integer is 8 more than twice the integer.

(A) 4

(B) 16

(C) 8

(D) 24

End of question paper. Maximum marks: 240. Marking: +4 correct, -1 wrong, 0 unattempted.

Solutions — Mathematics (Class 7), Chapter 15: Finding the Unknown

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	C	21	C	31	C	41	B	51	C
2	D	12	A	22	D	32	B	42	A	52	D
3	A	13	A	23	B	33	A	43	C	53	D
4	A	14	C	24	A	34	C	44	D	54	C
5	C	15	B	25	B	35	A	45	A	55	B
6	D	16	D	26	A	36	B	46	D	56	A
7	D	17	D	27	D	37	C	47	B	57	A
8	B	18	D	28	C	38	B	48	C	58	B
9	D	19	A	29	A	39	A	49	C	59	D
10	B	20	B	30	D	40	C	50	B	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) "Twice a number" = $2n$; "7 less than" it = $2n - 7$. ($2(n - 7)$ would be "7 less than the number, then doubled".)

2. (D) "Thrice a number" = $3x$; "5 added" = $3x + 5$; "result is 26" $\rightarrow 3x + 5 = 26$.

3. (A) $4x - 9 = 2x + 7 \rightarrow 4x - 2x = 7 + 9 \rightarrow 2x = 16 \rightarrow x = 8$. Then $3x + 1 = 3 \cdot 8 + 1 = 25$. (Stopping at $3x = 24$ gives the trap 24; using $2x + 1$ gives 17.)

4. (A) "Exceeds by 5" $\rightarrow x/3 - x/4 = 5$. Multiply by 12: $4x - 3x = 60 \rightarrow x = 60$. Check: $60/3 - 60/4 = 20 - 15 = 5$. ✓ (Reading "exceeds by 5" as $x/3 = x/4 + 5$ gives the same equation; the trap 5 is the difference itself.)

5. (C) $7x + 4 = 3x + 20 \rightarrow 7x - 3x = 20 - 4 \rightarrow 4x = 16 \rightarrow x = 4$. Check: $7 \cdot 4 + 4 = 32 = 3 \cdot 4 + 20$. ✓

6. (D) $x + 8 = 3(x - 4) \rightarrow x + 8 = 3x - 12 \rightarrow 8 + 12 = 3x - x \rightarrow 20 = 2x \rightarrow x = 10$. Check: $10 + 8 = 18 = 3(10 - 4) = 18$. ✓ (Forgetting to distribute the 3 over -4, i.e. $x + 8 = 3x - 4$, gives the trap 6.)

7. **(D)** $3(2x + 1) = 27 \rightarrow 6x + 3 = 27 \rightarrow 6x = 24 \rightarrow x = 4$. Check: $3(8 + 1) = 27$. ✓
8. **(B)** $(2x - 1)/3 = 5 \rightarrow 2x - 1 = 15 \rightarrow 2x = 16 \rightarrow x = 8$. Check: $(16 - 1)/3 = 5$. ✓
9. **(D)** $5x - 7 = 2x + 5 \rightarrow 3x = 12 \rightarrow x = 4$. Check: $20 - 7 = 13 = 8 + 5$. ✓
10. **(B)** Let the integers be $n, n + 1, n + 2$. Sum = $3n + 3 = 72 \rightarrow 3n = 69 \rightarrow n = 23$. Largest = $n + 2 = 25$. (23 is the smallest — a common trap.)
11. **(C)** Son = s , father = $4s$. $s + 4s = 50 \rightarrow 5s = 50 \rightarrow s = 10$. (40 is the father's age.)
12. **(A)** $4x + 10 = 38 \rightarrow 4x = 28 \rightarrow x = 7$.
13. **(A)** breadth = b , length = $b + 5$. Perimeter $2(b + 5 + b) = 50 \rightarrow 2(2b + 5) = 50 \rightarrow 2b + 5 = 25 \rightarrow 2b = 20 \rightarrow b = 10$. (Length is 15.)
14. **(C)** Sticks for n squares = $3n + 1$. Set $3n + 1 = 31 \rightarrow 3n = 30 \rightarrow n = 10$. Check: $3 \cdot 10 + 1 = 31$. ✓
15. **(B)** $x + 12 = 3x \rightarrow 12 = 2x \rightarrow x = 6$.
16. **(D)** $x/2 + x/3 = 10 \rightarrow$ multiply by 6: $3x + 2x = 60 \rightarrow 5x = 60 \rightarrow x = 12$. Check: $6 + 4 = 10$. ✓
17. **(D)** $4(x - 1) = 2(x + 3) \rightarrow 4x - 4 = 2x + 6 \rightarrow 2x = 10 \rightarrow x = 5$. Check: $4 \cdot 4 = 16 = 2 \cdot 8$. ✓
18. **(D)** Number = x , successor = $x + 1$. "Sum is 5 less than three times the number" $\rightarrow x + (x + 1) = 3x - 5 \rightarrow 2x + 1 = 3x - 5 \rightarrow 1 + 5 = 3x - 2x \rightarrow x = 6$. Check: $6 + 7 = 13 = 3 \cdot 6 - 5 = 13$. ✓
19. **(A)** breadth = b , length = $2b$. Perimeter = $2(2b + b) = 6b = 54 \rightarrow b = 9$. Length = $2b = 18$. Check: $2(18 + 9) = 54$. ✓ (9 is the breadth — the classic "answered the wrong unknown" trap.)
20. **(B)** Let t = number of ₹2 coins; then $20 - t$ are ₹1 coins. $2t + 1 \cdot (20 - t) = 35 \rightarrow t + 20 = 35 \rightarrow t = 15$. Check: $15 \cdot 2 + 5 \cdot 1 = 35$. ✓
21. **(C)** Test $x = 3$: (A) $2 \cdot 3 + 1 = 7 \neq 5$; (B) $3 - 3 = 0 \neq 3$; (C) $4 \cdot 3 - 5 = 7$ ✓; (D) $3/3 = 1 \neq 9$. So (C).
22. **(D)** "Half of a number, decreased by 6" = $x/2 - 6$; this equals one-third of it: $x/2 - 6 = x/3$. Multiply by 6: $3x - 36 = 2x \rightarrow x = 36$. Check: $36/2 - 6 = 12 = 36/3$. ✓
23. **(B)** $5 - 2(x - 1) = 1 \rightarrow 5 - 2x + 2 = 1 \rightarrow 7 - 2x = 1 \rightarrow 2x = 6 \rightarrow x = 3$. (The trap is writing $5 - 2x - 2$; the bracket flips the sign of -1 to $+2$.) Check: $5 - 2(2) = 1$. ✓

24. (A) Now: Sam = s , Reena = $s + 5$. In 3 years: $(s + 3) + (s + 5 + 3) = 33 \rightarrow 2s + 11 = 33 \rightarrow 2s = 22 \rightarrow s = 11$. Check: in 3 years $14 + 19 = 33$. ✓

25. (B) $2n + 1 = 25 \rightarrow 2n = 24 \rightarrow n = 12$. So the 12th term.

26. (A) $3(x + 2) = 2(x + 5) \rightarrow 3x + 6 = 2x + 10 \rightarrow x = 4$. Check: $3 \cdot 6 = 18 = 2 \cdot 9$. ✓

27. (D) $(x - 3)/2 = (x + 1)/4 \rightarrow$ cross-multiply: $4(x - 3) = 2(x + 1) \rightarrow 4x - 12 = 2x + 2 \rightarrow 2x = 14 \rightarrow x = 7$. Check: $4/2 = 2 = 8/4$. ✓

28. (C) $B = b$, $A = 2b$. $2b + b = 120 \rightarrow 3b = 120 \rightarrow b = 40$. (A gets 80.)

29. (A) $2x + 3 = 3x - 2 \rightarrow 3 + 2 = 3x - 2x \rightarrow x = 5$. Check: $2 \cdot 5 + 3 = 13 = 3 \cdot 5 - 2$. ✓

30. (D) $x + x + (x + 30) = 180 \rightarrow 3x + 30 = 180 \rightarrow 3x = 150 \rightarrow x = 50$. Each equal angle = 50° (third = 80°). ✓

31. (C) $12 - 3x = 2x - 8 \rightarrow 12 + 8 = 2x + 3x \rightarrow 20 = 5x \rightarrow x = 4$. Check: $12 - 12 = 0 = 8 - 8$. ✓

32. (B) $x/5 + 3 = 8 \rightarrow x/5 = 5 \rightarrow x = 25$.

33. (A) tens = t , units = $t + 3$. Digit-sum: $t + (t + 3) = 11 \rightarrow 2t + 3 = 11 \rightarrow 2t = 8 \rightarrow t = 4$, units = 7. The number is 47. Check: $4 + 7 = 11$ and $7 = 4 + 3$. ✓ (74 reverses the digits; 38 and 29 also sum to 11 but their units digit is not 3 more than the tens digit.)

34. (C) $2x/3 - x/2 = 1 \rightarrow$ multiply by 6: $4x - 3x = 6 \rightarrow x = 6$. Check: $12/3 - 6/2 = 4 - 3 = 1$. ✓

35. (A) Child tickets = c , adult = $5 - c$. $40(5 - c) + 25c = 155 \rightarrow 200 - 40c + 25c = 155 \rightarrow 200 - 15c = 155 \rightarrow 15c = 45 \rightarrow c = 3$. Check: 3 child (₹75) + 2 adult (₹80) = ₹155. ✓

36. (B) Test $x = -2$: (B) $2(-2) + 9 = 5$ ✓. The others: (A) $-5 \neq 5$, (C) $-4 \neq 0$, (D) $-8 \neq 8$. So (B).

37. (C) $2(3x - 1) = 4(x + 2) \rightarrow 6x - 2 = 4x + 8 \rightarrow 2x = 10 \rightarrow x = 5$. Check: $2 \cdot 14 = 28 = 4 \cdot 7$. ✓

38. (B) $50 + 12k = 158 \rightarrow 12k = 108 \rightarrow k = 9$.

39. (A) first = a , second = $2a$, third = $3a$. $a + 2a + 3a = 90 \rightarrow 6a = 90 \rightarrow a = 15$. Third = $3a = 45$. Check: $15 + 30 + 45 = 90$. ✓ (15 is the first friend's share, 30 the second — both are traps.)

40. (C) $(x - 1)/2 + (x + 1)/3 = 4$. Multiply by 6: $3(x - 1) + 2(x + 1) = 24 \rightarrow 3x - 3 + 2x + 2 = 24 \rightarrow 5x - 1 = 24 \rightarrow 5x = 25 \rightarrow x = 5$. Check: $4/2 + 6/3 = 2 + 2 = 4$. ✓ (Dropping a sign

on the brackets, e.g. $3x - 3 + 2x - 2$, gives $5x = 29$ and no clean answer — a signal you distributed wrongly.)

41. (B) $6x - 5 = 4x + 7 \rightarrow 2x = 12 \rightarrow x = 6$. Check: $36 - 5 = 31 = 24 + 7$. ✓

42. (A) Numbers x and $x + 6$. $x + (x + 6) = 30 \rightarrow 2x = 24 \rightarrow x = 12$. Larger = $x + 6 = 18$.

43. (C) $x + x + (x - 2) = 25 \rightarrow 3x - 2 = 25 \rightarrow 3x = 27 \rightarrow x = 9$. Check: $9 + 9 + 7 = 25$. ✓

44. (D) The terms go up by 3 from a first term of 5, so the n th term is $5 + 3(n - 1) = 3n + 2$. Set $3n + 2 = 47 \rightarrow 3n = 45 \rightarrow n = 15$. So 47 is the 15th term. Check: $3 \cdot 15 + 2 = 47$. ✓
(Picking 47 itself is the "the value IS the position" trap.)

45. (A) Cost of n notebooks = $15n$; "after a ₹20 discount" = $15n - 20$.

46. (D) units = u , tens = $2u$. Digit-sum $2u + u = 9 \rightarrow 3u = 9 \rightarrow u = 3$, tens = 6 $\rightarrow$ the number is 63. Check: $6 + 3 = 9$ and $6 = 2 \cdot 3$. ✓

47. (B) Test $x = 2$: (A) $5 = 5$ ✓, (B) $2 \cdot 2 + 3 = 7 \neq 5$ ✗, (C) $6 = 6$ ✓, (D) $10 = 10$ ✓. The exception is (B).

48. (C) $3 - (2x - 5) = 4 \rightarrow 3 - 2x + 5 = 4 \rightarrow 8 - 2x = 4 \rightarrow 2x = 4 \rightarrow x = 2$. Check: $3 - (4 - 5) = 3 - (-1) = 4$. ✓

49. (C) $A = a$, $B = a + 50$, $C = a + 100$. Sum = $3a + 150 = 600 \rightarrow 3a = 450 \rightarrow a = 150$.
Check: $150 + 200 + 250 = 600$. ✓

50. (B) $2x - 7 = x + 5 \rightarrow x = 12$. Check: $2 \cdot 12 - 7 = 17 = 12 + 5$. ✓

51. (C) Solve each and find the common value. First: $2x - 1 = x + 3 \rightarrow x = 4$. Second: $x/2 + 1 = 3 \rightarrow x/2 = 2 \rightarrow x = 4$. Both give $x = 4$, so 4 satisfies both. ✓ (A value satisfying only one equation — e.g. checking just the first — is the trap.)

52. (D) Angles x , $2x$, $3x$. $x + 2x + 3x = 180 \rightarrow 6x = 180 \rightarrow x = 30$. Largest = $3x = 90^\circ$.

53. (D) $x/2 + 4 = x - 1 \rightarrow 4 + 1 = x - x/2 \rightarrow 5 = x/2 \rightarrow x = 10$. Check: $10/2 + 4 = 9 = 10 - 1$. ✓

54. (C) $(x - 4)/3 = 5 \rightarrow x - 4 = 15 \rightarrow x = 19$.

55. (B) $n + (n+1) + (n+2) + (n+3) = 90 \rightarrow 4n + 6 = 90 \rightarrow 4n = 84 \rightarrow n = 21$. Check: $21 + 22 + 23 + 24 = 90$. ✓

56. (A) $2(x + 4) - 3 = x + 9 \rightarrow 2x + 8 - 3 = x + 9 \rightarrow 2x + 5 = x + 9 \rightarrow x = 4$. Check: $2 \cdot 8 - 3 = 13 = 4 + 9$. ✓

57. (A) daughter = d , mother = $d + 26$. In 6 years: $(d + 26 + 6) = 3(d + 6) \rightarrow d + 32 = 3d + 18 \rightarrow 14 = 2d \rightarrow d = 7$. Check: mother 33; in 6 years $39 = 3 \cdot 13$. ✓

58. (B) apples = a , oranges = $30 - a$. $8a + 5(30 - a) = 201 \rightarrow 8a + 150 - 5a = 201 \rightarrow 3a = 51 \rightarrow a = 17$. Check: $17 \cdot 8 + 13 \cdot 5 = 136 + 65 = 201$. ✓

59. (D) $(2x + 1)/3 = (x + 4)/2 \rightarrow$ cross-multiply: $2(2x + 1) = 3(x + 4) \rightarrow 4x + 2 = 3x + 12 \rightarrow x = 10$. Check: $21/3 = 7 = 14/2$. ✓

60. (C) $3x = 2x + 8 \rightarrow x = 8$. Check: $3 \cdot 8 = 24 = 2 \cdot 8 + 8$. ✓

Solutions complete: 60 / 60.